

Education

University of Tübingen, Tübingen, Germany

October 2025 – Present

- Master of Science in Machine Learning (German GPA 1.09, 1.00 = highest; WES U.S. equivalent: 4.00/4.00)
- Studies fully funded by ELIZA's Research-Oriented Master's Scholarship.

Middle East Technical University, Ankara, Turkey

October 2020 – July 2025

- Major in Statistics & Minor in Mathematics (CGPA: 3.91/4.00) – Rank 2/140

Research Experience

EPFL, NeuroAI Lab (Lausanne, Switzerland)

Research Intern (Advisors: Yingtian Tang, Professor Martin Schrimpf)

Aug 2026 – Present

- Developing causal intervention methods for controlling internal representations in vision-language models for computational neuroscience.
- Linking model representations and interventions to human neural responses, with the goal of identifying shared and divergent representations between artificial and biological systems.

University of Tübingen, CVML Group (Tübingen, Germany)

Research Assistant (Advisors: Edson Araujo, Professor Hilde Kuehne)

Apr 2026 – Present

- Developing JEPA-based multimodal alignment methods for self-supervised audio-visual representation learning.

Middle East Technical University (Ankara, Turkey)

Research Assistant (Advisors: Dr. Jiaee Cheong, Professor Sinan Kalkan)

Aug 2024 – Sep 2026

- Designed and implemented FairSSL, a self-supervised loss for modality-agnostic, fair multimodal representation learning tasks for neurological datasets (Joint work with Cambridge Affective Intelligence and Robotics Lab).
- Worked on model de-biasing strategies at the intersection of deep learning and kernel methods for fairness.

Undergraduate Researcher (Advisor: Professor Barbaros Yet)

Feb 2024 – Jan 2025

- Worked on causal agent-based models of decision theory in a medical shared decision-making setup.
- Modeled the conversation strategies employed by patients and doctors as a language game, consisting of various states, utilities, costs, and rewards.

Undergraduate Researcher (Advisor: Professor İlkey Ulusoy)

Jul 2023 – Jan 2024

- Applied Gaussian mixture models to inter-subject correlation features from fMRI to identify shared functional response regions.
- Modeled fMRI-derived functional connectivity using dynamic Bayesian networks to compare connectivity patterns between Alzheimer's disease and control groups.

Publications and Manuscripts

- **Abtin Mogharabin**, Jiaee Cheong, Sinan Kalkan (2026), From Point to Distributional Prediction in Joint-Embedding Predictive Architectures, *In preparation for ICLR 2027 Conference*
- **Abtin Mogharabin**, Jiaee Cheong, Alp Toykan Kaplan, Sinan Kalkan (2026), KINDER: Kernel-based Independence for Fair Representation Learning via Prototype-space Erasure, *Under review at NeurIPS 2026 Conference*
- Jiaee Cheong*, **Abtin Mogharabin***, Paul Liang, Hatice Gunes, Sinan Kalkan (2026), FairSSL: Fair multimodal self-supervised learning, *Accepted to ICML 2026 Proceedings* [PDF]
- **Abtin Mogharabin**, Vilda Purutçuoğlu. (2025), A Network Analysis of Family Dynamics, Linguistic Influences, and Happiness in Turkey, *Journal of Science: Engineering and Innovation, Gazi University, Part A, 12(3), 918-934* [PDF]
- **Abtin Mogharabin**. (2024), Modeling Emotional Functions in Medical Shared Decision-Making, preprint [PDF]

Teaching

- Student assistant for the Mathematical Statistics I and II courses (Instructor: Professor Zeynep Kalaylıoğlu) during the 2022-2023 academic year. Offered office hours for over 100 students and provided weekly individualized support and problem-solving.
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Selected Technical Projects

- Extended the experimental variable elimination library of Gen.jl probabilistic programming system (GenVariableElimination.jl) with a JAX einsum backend for scalable factor-graph contraction on GPUs, preserving modeling interface and semantics across large models with over 10x speedup on specific tests. [[Repo](#)]
 - Implemented and reproduced the results of Bayesian Normalization Layers (BNL), CVPR 2024, contributing to the field as no implementation was available for this paper. [[Repo](#)]
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Skills and Interests

- **Programming Skills:** Proficient in Python, R, and Julia. Prior experience with C++, JavaScript, MATLAB, and Haskell
 - **Machine Learning Tools:** Proficient in PyTorch, JAX, NumPy, scikit-learn, PyMC.
 - **Miscellaneous:** Proficient in Linux/UNIX, Git, L^AT_EX, Slurm
 - **Languages:** Persian (Native), English (Full Professional Proficiency)
 - **Research Interests:** Computational Neuroscience, NeuroAI, Representation Learning, Causal Interventions, Trustworthy AI
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Awards and Achievements

- Accepted to EPFL's Summer@EPFL Research Internship Program (2026) at the NeuroAI lab with Professor Martin Schrimpf, selected from ~8,000 applicants (1.2% acceptance rate).
- ELIZA Excellence Research-Based Master's Scholarship: Master's studies at Tübingen fully funded by Zuse School of Excellence in Learning and Intelligent Systems (ELIZA).
- Achieved the second highest cumulative GPA in the batch of 140 graduating statistics students at METU (2025).
- METU Statistics High Honors Roll, Dean's List (2020-2025).
- Received METU's 100% international students' tuition fee payment scholarship (2021-2024).